



Assignment - 01

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Submitted To

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Table of Contents

1. Introduction	2
2.1 Programming Environment	3
2.2 Libraries Used	3
2.3 Installation on Kaggle.....	3
3.1 YouTube Video Selection.....	3
3.2 Captions Collection.....	4
3.3 Comments Collection.....	4
4.1 Loading the Data	4
4.2 Cleaning and Preprocessing.....	6
4.3 Purpose of Preprocessing	7
5. Experiments and Results	7
5.1 Caption VS Comment Length Distribution	7
5.2 Vocabulary Diversity (Type-Token Ratio).....	8
5.3 Comparative Analysis of Vocabulary Diversity in Captions and Comments ...	9
5.4 Boxplot of Text Lengths	9
5.5 Top 20 Caption Words	10
5.6 Top 20 Comment Words	11
5.7 Smoothed Caption Length Trend (Moving Average).....	12
6. Analysis and Discussion	13
6.1 Challenges Faced.....	13
7. Conclusion	14

1. Introduction

The goal of this lab is to perform Exploratory Data Analysis (EDA) on textual data collected from YouTube. Specifically, this lab focuses on comparing YouTube video captions and user comments.

Captions usually represent structured, spoken content from the video itself, while comments are user-generated and often informal, emotional and diverse. Through this lab, we aim to explore differences between captions and comments in terms of:

- Text length distribution
- Vocabulary diversity
- Word frequency patterns

This analysis helps in understanding how different types of textual data behave and how they can be used in Natural Language Processing (NLP) tasks.

2. Tools and Setup

2.1 Programming Environment

All experiments in this lab were conducted using a Kaggle Notebook environment. Kaggle provides a cloud-based Python platform with pre-installed libraries, making it suitable for reproducible and efficient data analysis.

- **Platform:** Kaggle Notebook
- **Python Version:** Python 3.10

2.2 Libraries Used

The following Python libraries were used throughout the lab:

- **pandas**
Used for reading text files, storing data in DataFrames and performing data manipulation and statistical analysis.
- **matplotlib**
Used to create visualizations such as histograms, bar charts and box plots for comparative analysis.
- **collections.Counter**
Used to calculate word frequency distributions and identify the most frequent words after preprocessing.

- **yt-dlp**

Used to download YouTube video captions in WebVTT (.vtt) format.

2.3 Installation on Kaggle

The **yt-dlp** library is not included by default in Kaggle. Therefore, it was installed directly within the notebook using the following command:

```
!pip install -q yt-dlp
```

All experiments, including data collection, preprocessing, analysis and visualization, were performed entirely within the Kaggle environment.

3. Data Collection

3.1 YouTube Video Selection

- **Video URL:** <https://youtu.be/h3M00JI8Iwo?si=cPmNca1zhn7voz19>
- The video was selected according to the criteria provided in the lab manual.
- It is a publicly accessible YouTube video with English captions available, which is necessary for caption analysis.
- The video also contains enough user comments, making it suitable for comparative analysis between captions and comments.

3.2 Captions Collection

- Video captions were downloaded using the **yt-dlp** tool within the Kaggle Notebook environment.
- Automatic English captions were extracted and saved in **WebVTT (.vtt)** format.
- The captions file was renamed to **captions.vtt** for consistency and ease of access.
- The .vtt file was uploaded to the Kaggle dataset section and accessed using a **relative file path** inside the notebook.
- This approach ensured that caption data could be processed directly without relying on any external services during runtime.

3.3 Comments Collection

- User comments associated with the selected YouTube video were collected and stored in a text file named **Comments.txt**.
- Each line in the file corresponds to a single YouTube comment.
- The comments file was uploaded to the Kaggle dataset and read directly within the notebook using standard file handling methods.

- Empty or irrelevant lines were ignored during preprocessing to maintain data quality.

No external APIs were used during runtime. All data files, including captions and comments, were handled locally within the Kaggle environment, ensuring a controlled and reproducible data collection process.

4. Data Processing

4.1 Loading the Data

- **Captions:** The captions were stored in a `.vtt` file, which contains both the spoken text and timestamps indicating when each line appears in the video. The file was read line by line, extracting only the text content while ignoring the timestamp and formatting information.
- **Comments:** User comments were stored in a `.txt` file, with each comment typically on a separate line. These were loaded as plain text into the analysis environment.

This step ensures that both captions and comments are imported and ready for preprocessing and analysis.

Code:

```
def load_vtt_captions(filepath="captions.vtt"):
    captions = []
    with open(filepath, "r", encoding="utf-8") as f:
        for line in f:
            line = line.strip()
            if (
                not line
                or "WEBVTT" in line
                or "-->" in line
                or line.isdigit()
                or "Kind:" in line
                or "Language:" in line
            ):
                continue
```

```

# Remove inline timestamps like <00:00:05.920>
while "<" in line and ">" in line:
    start = line.find("<")
    end = line.find(">", start)
    if end != -1:
        line = line[:start] + line[end + 1:]
    else:
        break
line = line.strip()
if line:
    captions.append(line)

return captions

raw_captions = load_vtt_captions("/kaggle/working/captions.vtt")
print(f'Loaded {len(raw_captions)} caption lines')
print("Sample captions:", raw_captions[:5])

```

4.2 Cleaning and Preprocessing

Raw text data often contains noise such as timestamps, special characters or irrelevant words. To make the data suitable for analysis, the following preprocessing steps were applied:

1. **Removal of Timestamps, Metadata and Empty Lines (Captions):**
Timestamps (e.g., 00:01:23.000 to 00:01:25.000) and other formatting metadata were removed. Empty lines, which do not contain any meaningful text, were also discarded. This ensures only the actual spoken words remain for analysis.
2. **Conversion to Lowercase:**
All text was converted to lowercase. This standardizes the data so that words like “Hello” and “hello” are treated as the same word, preventing duplication in analysis like word frequency counts.
3. **Removal of Unnecessary Symbols and Stopwords:**
Special characters, punctuation marks and irrelevant symbols were removed. Additionally, common words that carry little semantic meaning (e.g., “the”, “and”, “is”) were removed during word frequency analysis. This helps focus on meaningful content.
4. **Calculation of Text Length:**
The length of each caption or comment was measured based on the number of

characters. This can help in understanding text verbosity or for filtering extremely short or long entries if needed.

Code:

```
def load_raw_comments(filepath):  
    comments = []  
  
    with open(filepath, "r", encoding="utf-8") as f:  
        for line in f:  
            line = line.strip()  
            if len(line) > 2:  
                comments.append(line)  
  
    return comments  
  
# Replace 'YOUR_DATASET' with your Kaggle dataset folder name  
raw_comments = load_raw_comments("/kaggle/input/commments/Comments.txt")  
  
print(f"Loaded {len(raw_comments)} comment lines")  
  
print("Sample comments:", raw_comments[:5])
```

4.3 Purpose of Preprocessing

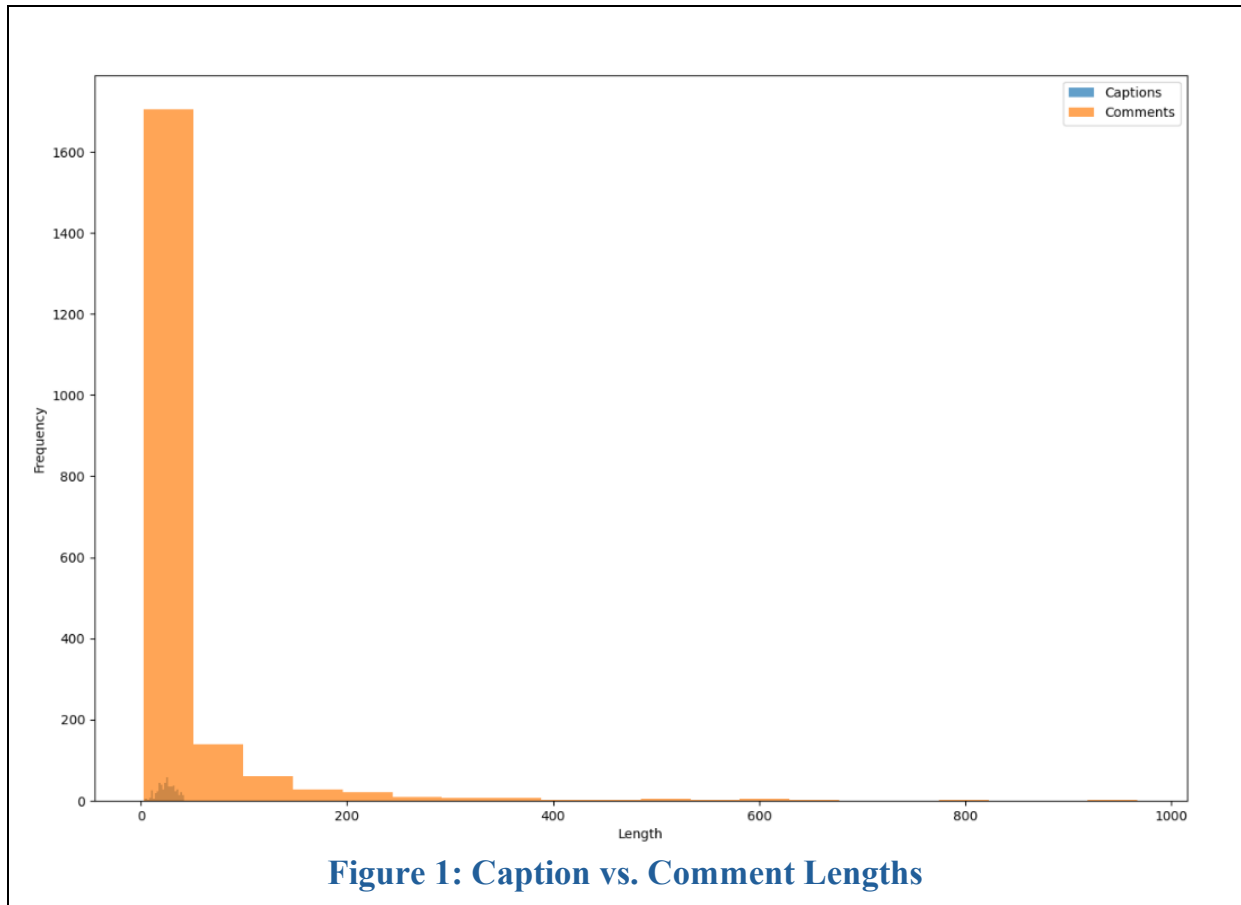
These preprocessing steps ensure that the text data is clean, consistent and ready for analysis. Proper preprocessing improves the accuracy of further analyses, such as:

- Word frequency analysis
- Sentiment analysis
- Keyword extraction
- Visualization of text trends

5. Experiments and Results

In this I will analyze and discuss the experiments and visualization I performed on my created YouTube dataset. It will also include the discussion about the challenges I faced during the whole process.

5.1 Caption VS Comment Length Distribution



The histogram shows the distribution of text lengths (in characters) for captions and comments. Captions (blue) are generally short and consistent, mostly under 50 characters, reflecting concise spoken text in videos. Comments (orange) vary widely: many are very short, but a few extend beyond 900 characters, creating a long-tailed distribution. This contrast highlights that captions are structured and predictable, while comments are diverse, which is important to consider for text analysis tasks like sentiment detection or keyword extraction.

5.2 Vocabulary Diversity (Type-Token Ratio)

Vocabulary diversity was measured using the Type-Token Ratio (TTR), defined as:

$$\text{TTR} = \text{Number of Unique Words} / \text{Total Number of Words}$$

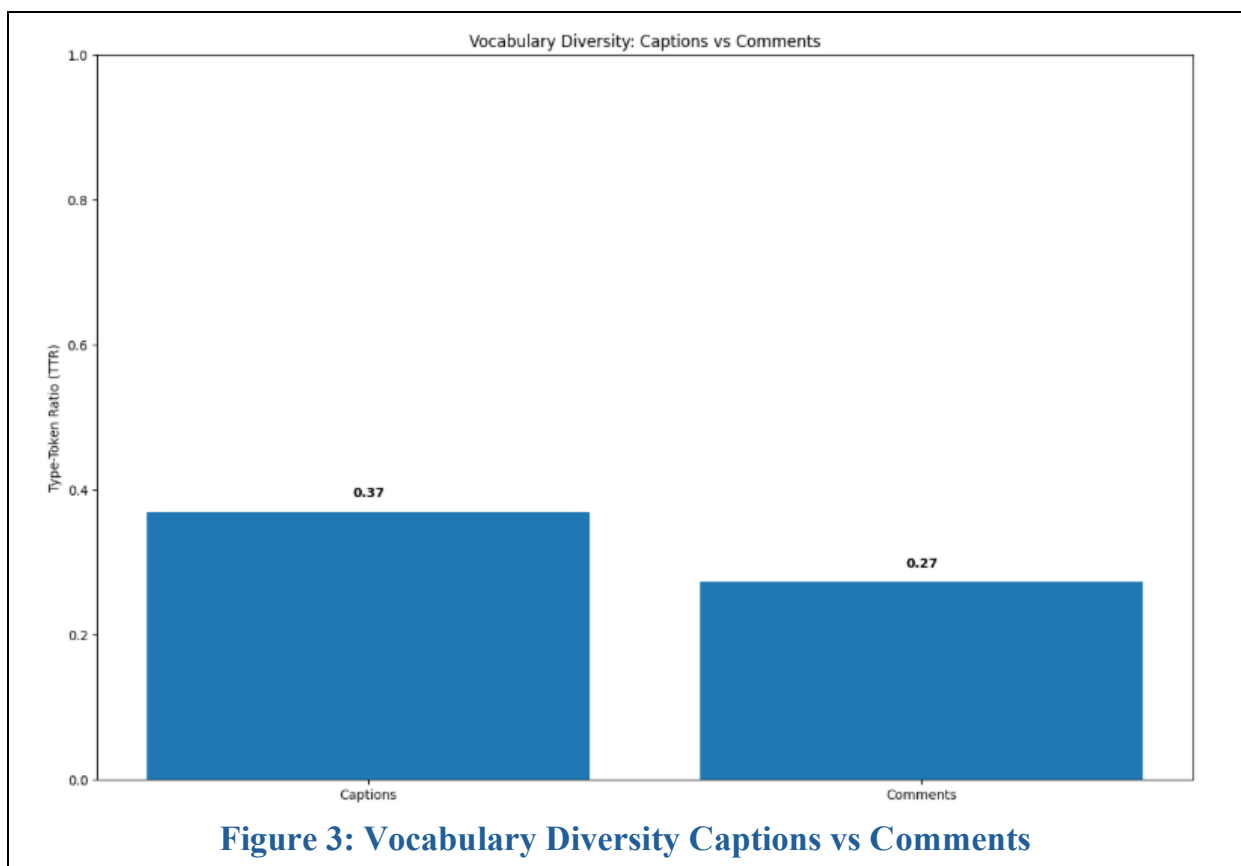
Result:

- Captions exhibit a lower TTR, indicating repetitive and topic-focused language.
- Comments show a higher TTR, reflecting diverse vocabulary usage.

Caption TTR: 0.3695652173913043
Comment TTR: 0.2725375988872468

Figure 2: Type-Token Ratio

5.3 Comparative Analysis of Vocabulary Diversity in Captions and Comments



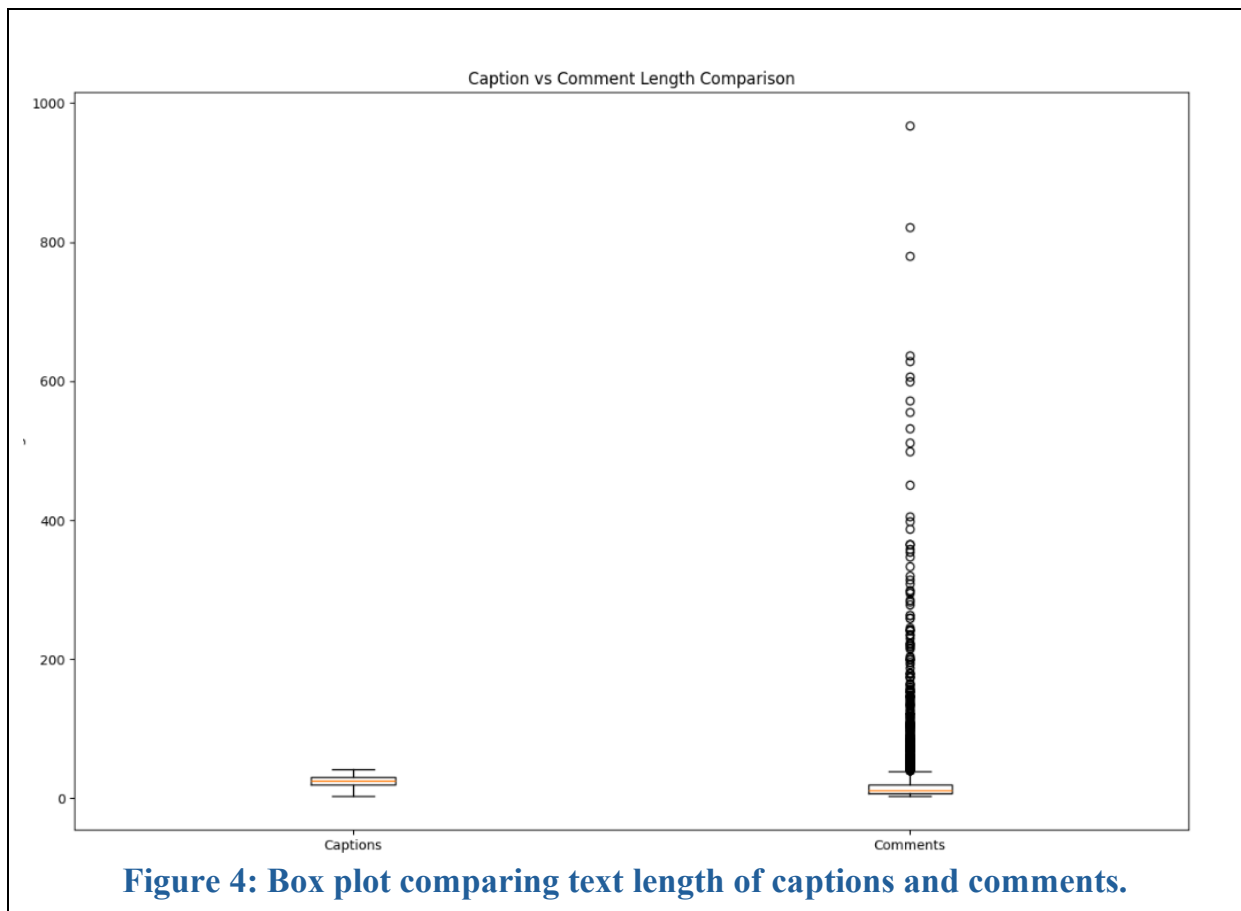
The graph presents a comparison of Type–Token Ratio (TTR) between captions and comments to analyze vocabulary diversity. TTR represents the proportion of unique words relative to the total number of words in a text. The captions have a higher TTR value (0.37) compared to comments (0.27), indicating that captions use a wider and more varied vocabulary. On the other hand, the lower TTR for comments suggests more frequent repetition of words. This shows that captions are generally more expressive, while comments tend to be simpler and more repetitive in language use.

5.4 Boxplot of Text Lengths

A boxplot was generated to compare caption and comment lengths.

Result:

- Comments contain more outliers.
- Caption lengths are more evenly distributed.



The box plot compares the length of captions and comments. Captions show a narrower range, meaning most are short and consistent, with fewer extreme cases. In contrast, comments display a much wider spread, including many outliers, which indicates that users often write both very short and very long comments.

Overall, this suggests that captions are usually concise and uniform, while comments vary greatly in length, reflecting different levels of user engagement.

5.5 Top 20 Caption Words

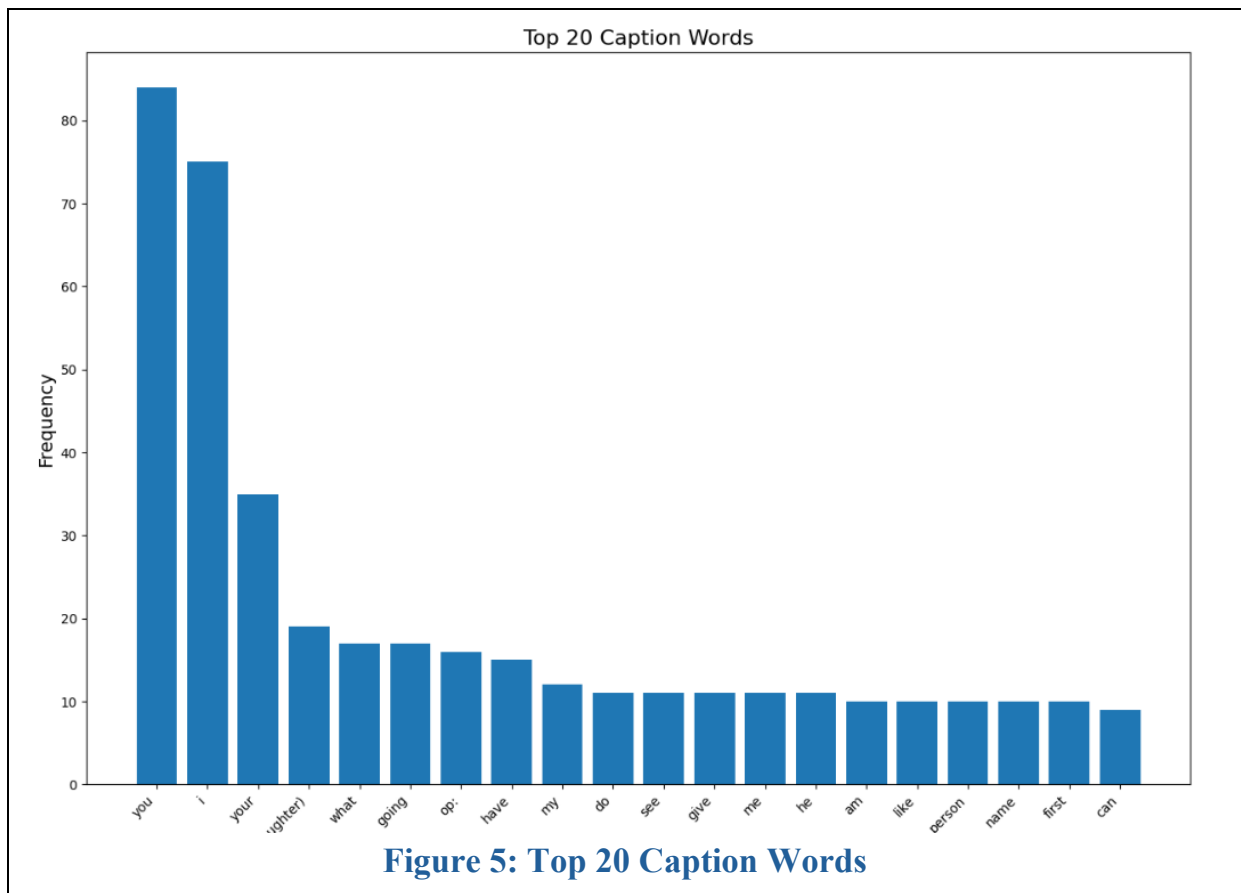
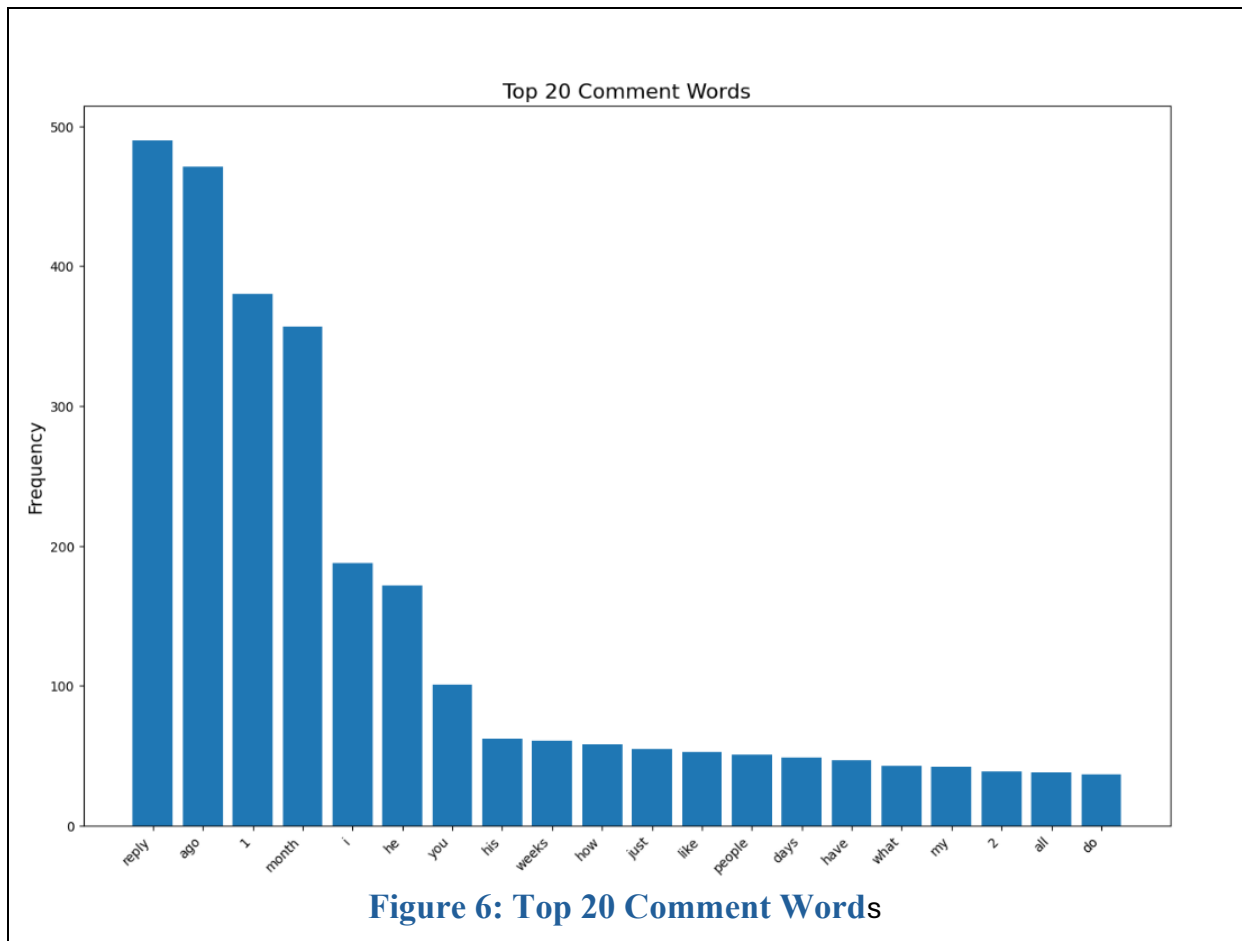


Figure 5: Top 20 Caption Words

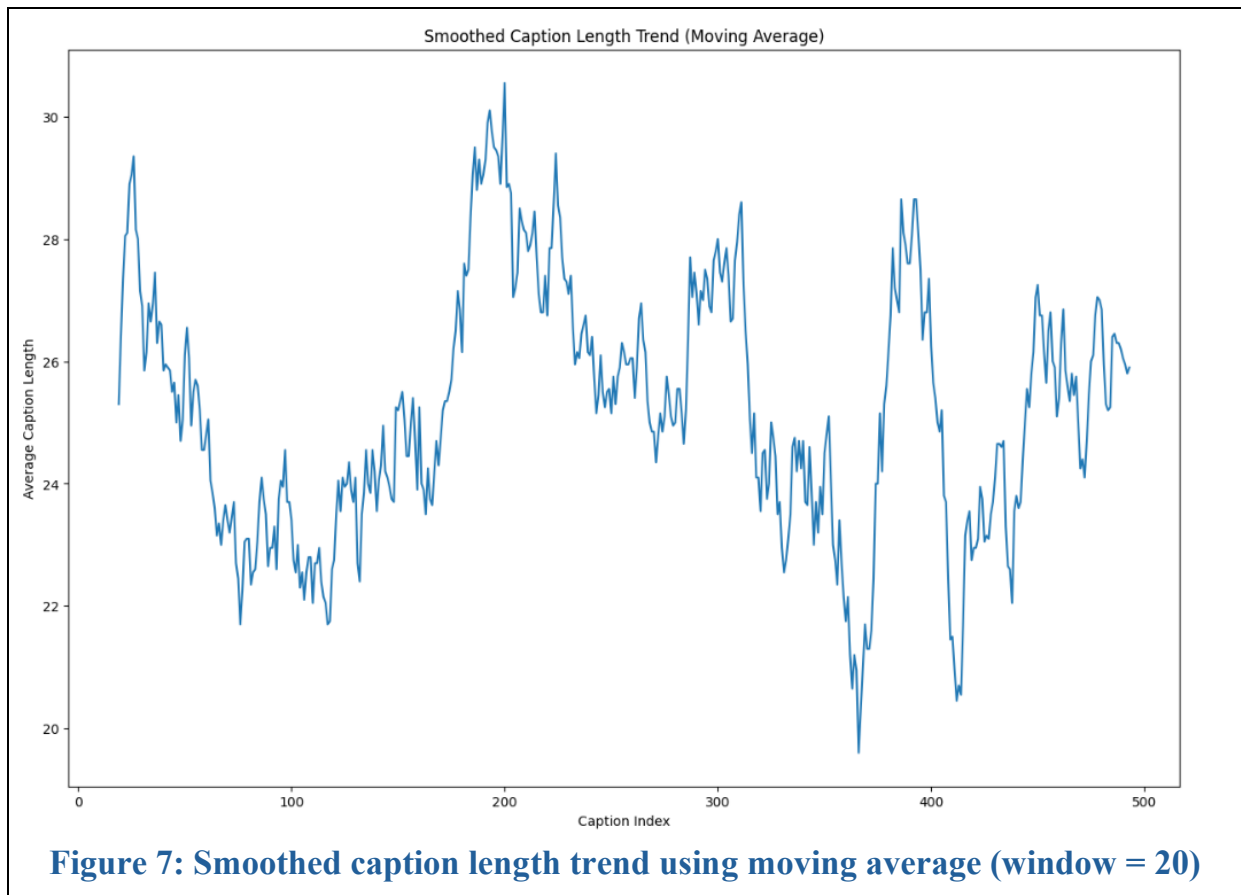
This bar chart displays the 20 most frequently used words in captions. The most common words “you,” “I,” “your,” and “daughter” appear with noticeably higher frequencies, suggesting a personal and relational tone in caption writing. After the top few, the frequency drops sharply, and the remaining words show relatively similar usage levels. This pattern indicates that captions often rely on a small set of emotionally expressive or direct-address words, reflecting their role in conveying personal messages or storytelling.

5.6 Top 20 Comment Words



This chart shows the most frequent words found in comments. The top words “reply,” “ago,” “1” and “month” are significantly more frequent than the rest, pointing to a strong presence of time-related and interaction-based language. Unlike captions, comment vocabulary includes more references to time, responses, and social engagement. The gradual decline in word frequency across the rest of the chart suggests a broader spread of commonly used terms, reflecting the diverse nature of user interactions in comment sections.

5.7 Smoothed Caption Length Trend (Moving Average)



This line graph illustrates how caption lengths vary across the dataset. Each point represents the moving average of caption length over a window of 20 captions, helping smooth out short-term fluctuations and highlight overall trends. The x-axis shows the caption index, while the y-axis indicates the average length.

The plot reveals subtle variations in caption length, suggesting that while most captions maintain a consistent range, there are periodic shifts possibly reflecting changes in content style or speaker behavior over time.

6. Analysis and Discussion

The experimental results highlight clear differences between captions and comments:

- Captions are structured, concise, and less diverse.
- Comments are informal, expressive, and lexically diverse.
- User-generated text shows greater variability in both length and vocabulary.

6.1 Challenges Faced

- Cleaning .vtt files required careful removal of timestamps and metadata.
- Some comments contained noise such as emojis and irregular characters.

Despite these challenges, meaningful insights were successfully extracted from the data.

7. Conclusion

In this lab, exploratory data analysis was performed on YouTube captions and comments using a Kaggle Notebook environment. The analysis revealed significant linguistic and structural differences between the two text sources. Captions were found to be concise and repetitive, while comments were more diverse and expressive.

This lab provided hands on experience with real world textual data and established a strong foundation for future NLP tasks such as sentiment analysis, topic modelling and text classification.